Teaching notes and solutions

Teacher notes

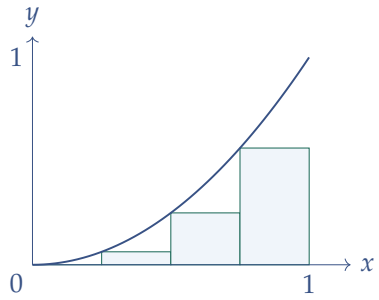
Begin with a positive area, then show an example in which the positive and negative areas cancel. For a mean value, require a unit and an interpretation. Sequences of integrals arrive in S66; reserve a genuine research phase for the choice of comparison. The proof of the fundamental theorem is limited to the positive increasing case stated in the syllabus. Trigonometric calculations will be taken up again after May.

Solutions to the activity

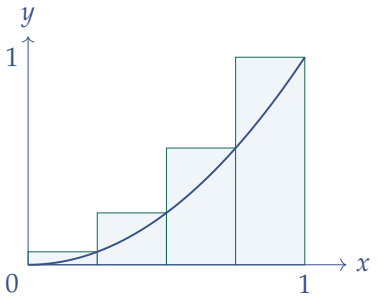
1. The four intervals have width $\frac{1}{4}$. The left sum is

$$L_4 = \frac{1}{4}(0^2 + (\frac{1}{4})^2 + (\frac{1}{2})^2 + (\frac{3}{4})^2) = \frac{7}{32}.$$

On the right, $U_4 = \frac{1}{4}((\frac{1}{4})^2 + (\frac{1}{2})^2 + (\frac{3}{4})^2 + 1) = \frac{15}{32}$. The rectangles have heights equal to the values at the corresponding endpoints.



Left rectangles: $L_4 = \frac{7}{32}$.



Right rectangles: $U_4 = \frac{15}{32}$.

2. For $x \in [\frac{k}{4}, \frac{k+1}{4}]$, $(\frac{k}{4})^2 \le x^2 \le (\frac{k+1}{4})^2$ because the square function is increasing on $[0, 1]$. Integrating over the four intervals then adding gives $\frac{7}{32} \le I \le \frac{15}{32}$.

3.
```
def rectangles_carre(n):
    h = 1/n
    bas = h*sum((k*h)**2 for k in range(n))
    haut = h*sum((k*h)**2 for k in range(1, n+1))
    return bas, haut
```

The inner terms cancel: $U_n - L_n = (\frac{1}{n})(1^2 - 0^2) = \frac{1}{n}$.

4. $\frac{1}{n} \le 0.01 \iff n \ge 100$. One hundred subdivisions suffice; this is the first integer that gives this width with these two sums.

5. An antiderivative of x^2 is $\frac{x^3}{3}$. So $I = [\frac{x^3}{3}]_0^1 = \frac{1}{3}$, which indeed lies between $\frac{7}{32}$ and $\frac{15}{32}$.

E1. An antiderivative is $F(x) = x^3 - x^2 + x$. The integral is $F(2) - F(0) = 8 - 4 + 2 = 6$.

E2. On $[0, 1]$, $x - x^2 = x(1 - x) \ge 0$, so the line is above the parabola. The area is $\int_0^1 (x - x^2)\,\mathrm{d}x = [\frac{x^2}{2} - \frac{x^3}{3}]_0^1 = \frac{1}{6}$ area unit.

E3. The length of the interval is 2. The mean value is $(\frac{1}{2})\int_0^2 e^x\,\mathrm{d}x = (\frac{1}{2})[e^x]_0^2 = \frac{e^2-1}{2}$.

E4. Choose $u = \ln x$, $u' = \frac{1}{x}$, $v' = 1$ and $v = x$ on $[1, e]$. Integration by parts gives $\int_1^e \ln x \, dx = [x \ln x]_1^e - \int_1^e 1 \, dx = e - (e-1) = 1$.

E5. On $[0,1]$, $1 + x \geq 1$ and $x^n \geq 0$, so $0 \leq \frac{x^n}{1+x} \leq x^n$. Integrating gives $0 \leq I_n \leq [\frac{x^{n+1}}{n+1}]_0^1 = \frac{1}{n+1}$. The squeeze theorem gives $I_n \to 0$.

Solutions to HW8

1. $I_0 = \int_0^1 e^x \, dx = e - 1$. For I_1, integration by parts with $u = x$, $v' = e^x$ gives $I_1 = [xe^x]_0^1 - \int_0^1 e^x \, dx = e - (e-1) = 1$.
2. For $n \geq 0$, choose $u = x^{n+1}$, $u' = (n+1)x^n$, $v' = e^x$ and $v = e^x$. The functions are differentiable with continuous derivatives on $[0,1]$. Then
$$I_{n+1} = [x^{n+1}e^x]_0^1 - (n+1)\int_0^1 x^n e^x \, dx = e - (n+1)I_n.$$
3. On $[0,1]$, $0 \leq x^{n+1} \leq x^n$. Multiplying by $e^x > 0$ then integrating, $0 \leq I_{n+1} \leq I_n$: the sequence is positive and non-increasing. It is even strictly positive and strictly decreasing, because the integrands of I_n and of $I_n - I_{n+1}$ are strictly positive on $]0,1[$.
4. Since $1 \leq e^x \leq e$ and $x^n \geq 0$ on $[0,1]$, we can integrate $x^n \leq x^n e^x \leq e x^n$. Hence
$$\frac{1}{n+1} \leq I_n \leq \frac{e}{n+1}.$$
Both bounds tend to 0; the squeeze theorem gives $I_n \to 0$.
5. If $\widetilde{I}_n = I_n + \varepsilon_n$ and the relation is then applied with no new error, then
$$\widetilde{I}_{n+1} - I_{n+1} = -(n+1)(\widetilde{I}_n - I_n) = -(n+1)\varepsilon_n.$$
The absolute value of the error is multiplied by $n+1$. In machine calculation, new errors are added at each step. The decrease of the exact integrals therefore does not guarantee the stability of this method of calculation.

Solutions to the textbook or past-paper work

Exercise 28. On $[0;1]$, choose $u(x) = x$ and $v(x) = e^x$. These functions are of class $\mathcal{C}^1$, $u'(x) = 1$ and $v'(x) = e^x$. Integration by parts gives
$$\int_0^1 xe^x \, dx = [xe^x]_0^1 - \int_0^1 e^x \, dx = e - (e-1) = 1.$$

Exercise 29. On $[1;e]$, choose $u(x) = \ln x$ and $v(x) = \frac{x^2}{2}$. Then $u'(x) = \frac{1}{x}$ and $v'(x) = x$; these functions are of class $\mathcal{C}^1$. Hence
$$\begin{aligned}\int_1^e x \ln x \, dx &= \left[\frac{x^2}{2}\ln x\right]_1^e - \frac{1}{2}\int_1^e x \, dx \\ &= \frac{e^2}{2} - \frac{e^2-1}{4} = \frac{e^2+1}{4}.\end{aligned}$$

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